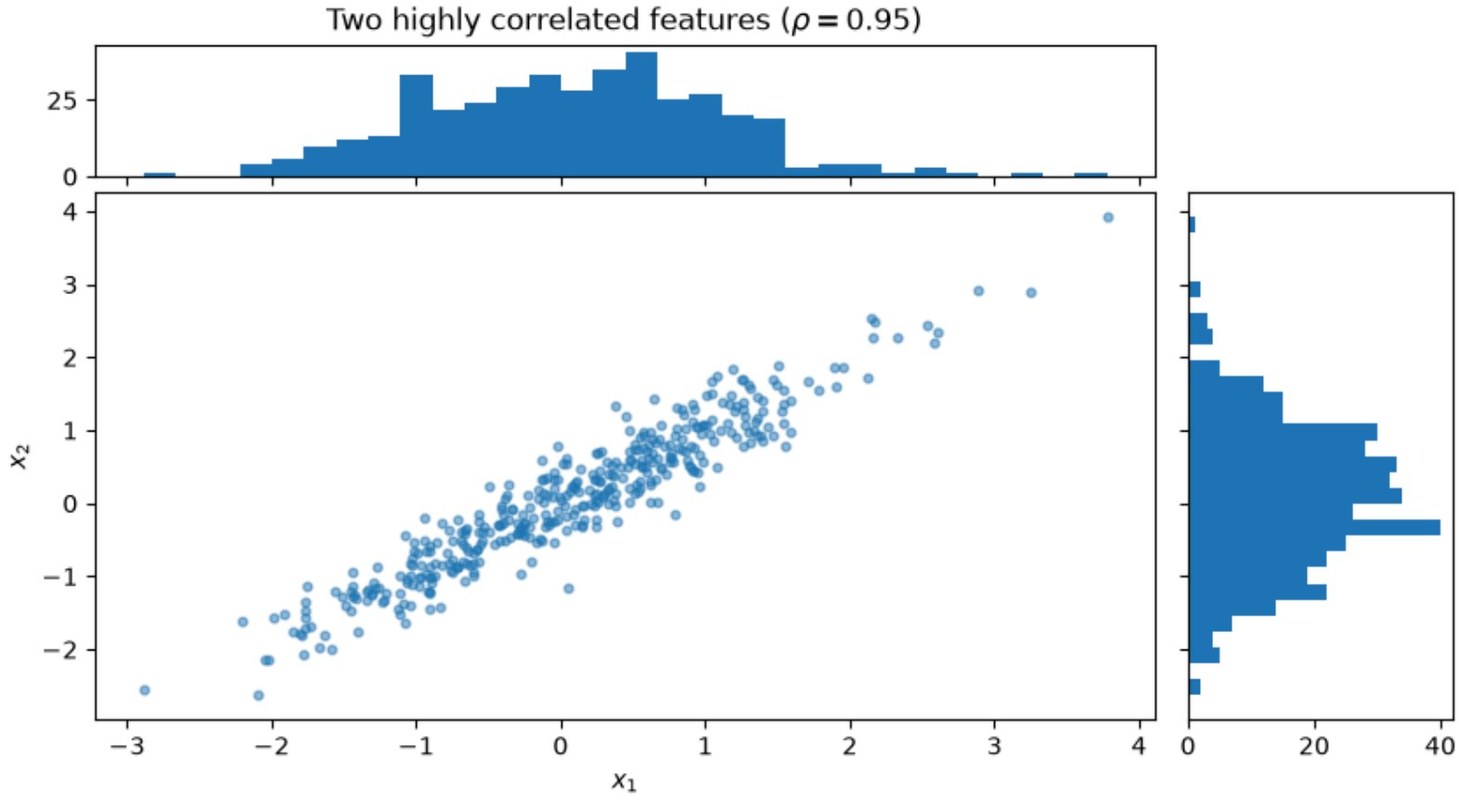
Data transformation and dimensionality reduction (DTDR) I

Introduction

- So far, the course has been problem oriented (mainly classification).
- However, an essential part of pattern recognition is data analysis.
- In this lecture, we will look at methods that compress data into a more compact representation through linear transformations.

Analyzing data

- Real data often live in a high-dimensional space, but the information they carry may be concentrated on a much smaller subspace.
- Example below: two features that are almost perfectly correlated.
    - Little is lost by describing the cloud with a single direction.


Two highly correlated features ($\rho = 0.95$)
25
0
4
3
2
1
0
−1
−2
x_2
−3
−2
−1
0
1
2
3
4
x_1
0
20
40

Problems

- The data dimensionality may be very large.
    - Computationally demanding.
    - Curse of dimensionality
- Some parts of the data may not be discriminative / be redundant.

Remedies

- May pick only parts of the data to use.
- May transform the data!

Approaches to DTDR

- To better discriminate between classes (supervised).
- To remove redundancy (unsupervised).

Fisher discriminant analysis (FDA)

- Transform (project) to 1D: $z = \mathbf{w}^T \mathbf{x}$
- Start with the two class case and $P(w_1) = P(w_2)$
- Fisher discriminant ratio (FDR) — large is good:

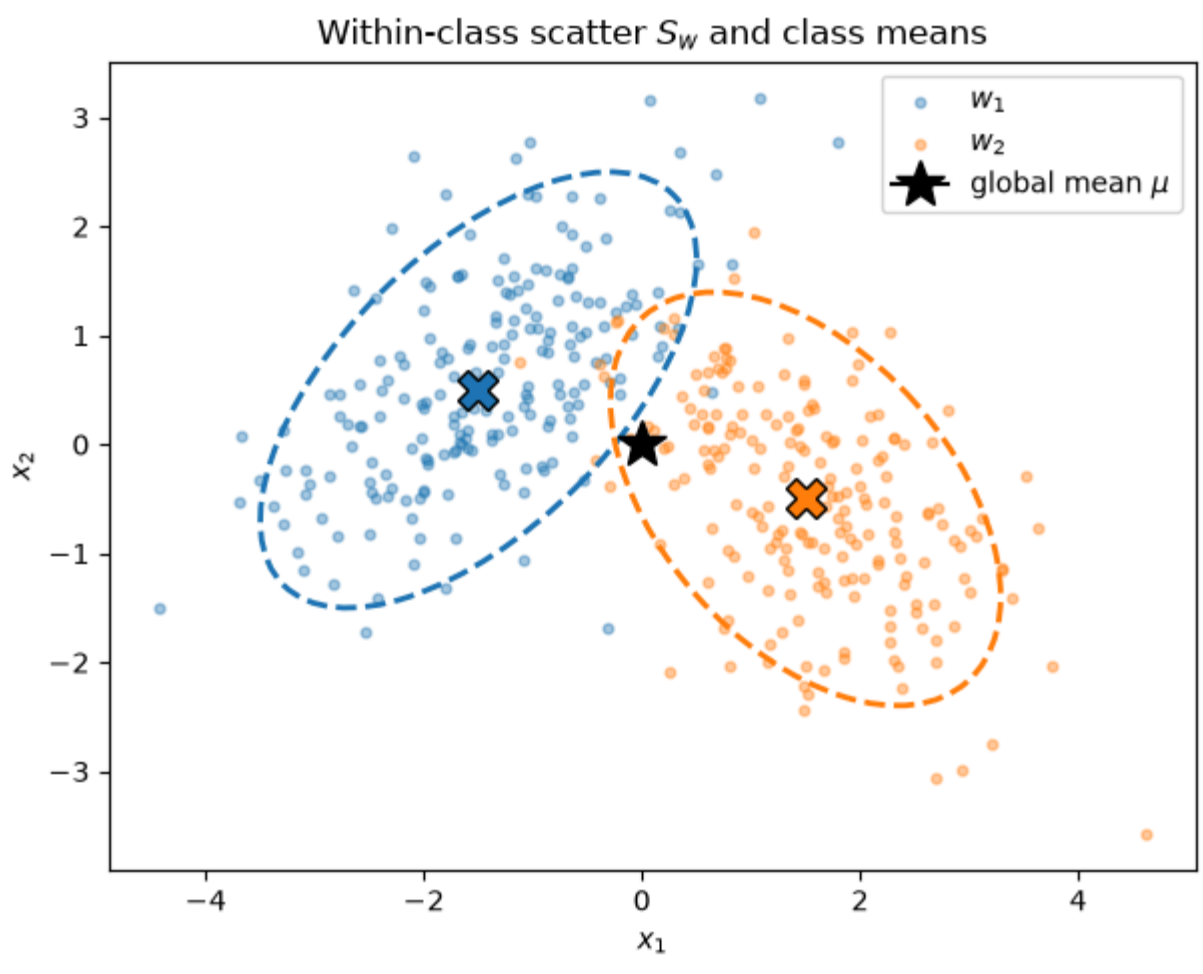
$$\mathrm{FDR} = \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2}$$

where μ_i and σ_i^2 are the mean and variance of the projected class-i data.

Scatter matrices

- Within class:
  $\boldsymbol{S}_w = \sum_{i=1}^{M} P(w_i)\boldsymbol{\Sigma}_i$
- where
  $\boldsymbol{\Sigma}_i = \mathbb{E}\left[(\mathbf{x} - \boldsymbol{\mu}_i)(\mathbf{x} - \boldsymbol{\mu}_i)^T\right]$

- Between class: $\boldsymbol{S}_B = \sum_{i=1}^{M} P(w_i)(\boldsymbol{\mu}_i - \boldsymbol{\mu})(\boldsymbol{\mu}_i - \boldsymbol{\mu})^T$



, $\boldsymbol{S}_B$ spread *between*

Remark

- Class separability in $\mathbf{x}$ can be measured by e.g.

$$\frac{\operatorname{trace}(\boldsymbol{S}_w)}{\operatorname{trace}(\boldsymbol{S}_B)}$$

- Here **small is good**: little within-class spread relative to the between-class spread.

Fisher discriminant analysis

- Remember; want to learn a transformation into 1D $z = \mathbf{w}^T \mathbf{x} \;\Rightarrow\; \mu_z = \mathbf{w}^T \boldsymbol{\mu}$
- In the projected space, with equal priors $P(w_1) = P(w_2) = \frac{1}{2}$:

$$S_B = (\mu_1 - \mu_2)^2, \qquad \mu_i = \mathbf{w}^T \boldsymbol{\mu}_i$$

$$\sigma_i^2 = \mathbb{E}\left[(z - \mu_i)^2\right] = \mathbf{w}^T \boldsymbol{\Sigma}_i \mathbf{w}$$

$$S_w = \tfrac{1}{2}\sigma_1^2 + \tfrac{1}{2}\sigma_2^2$$

- Hence: Fisher discriminant ratio (FDR) — to be **maximized**:

$$\mathrm{FDR}(\mathbf{w}) = \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2}$$

FDR

- Want:

$$\arg\max_{\mathbf{w}} \frac{\mathbf{w}^T \boldsymbol{S}_B \mathbf{w}}{\mathbf{w}^T \boldsymbol{S}_w \mathbf{w}}$$

- At the solution (generalized eigenvalue problem): $\boldsymbol{S}_B \mathbf{w} = \lambda \boldsymbol{S}_w \mathbf{w} \Leftrightarrow \boldsymbol{S}_w^{-1} \boldsymbol{S}_B \mathbf{w} = \lambda \mathbf{w}$

Eigenvalue problems

- For a square matrix $\mathbf{A}$, an **eigenvector** $\mathbf{v} \neq \mathbf{0}$ and its **eigenvalue** λ satisfy

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

- $\mathbf{A}$ stretches $\mathbf{v}$ by λ without changing its direction.

- For the covariance matrix $\mathbf{\Sigma}$ (symmetric, positive semi-definite):

$$\mathbf{\Sigma}\mathbf{e}_i = \lambda_i \mathbf{e}_i, \qquad \lambda_i \geq 0$$

- The eigenvectors are orthogonal and the eigenvalue λ_i is the **variance** of the data along $\mathbf{e}_i$.
- The largest eigenvalue gives the **direction of greatest variance** (the first principal component).


Eigenvectors of Σ: directions of greatest variance
e_2 $\lambda_2 = 0.14$
e_1 $\lambda_1 = 3.06$
x_1
x_2
3
2
1
0
−1
−2
−3
−2
0
2
4

Solving the Fisher discriminant ratio

I. If $P(w_1) = P(w_2)$:

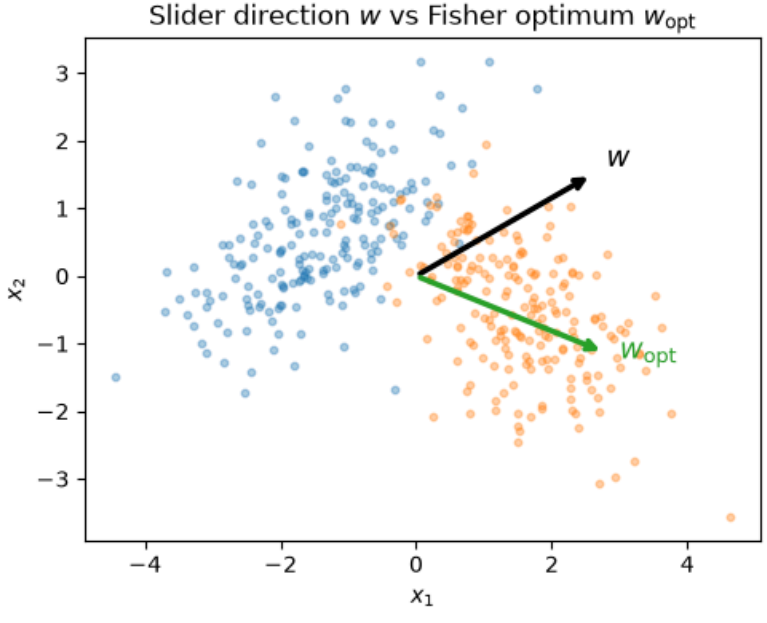
$$\lambda \boldsymbol{S}_w \mathbf{w} = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \mathbf{w}$$

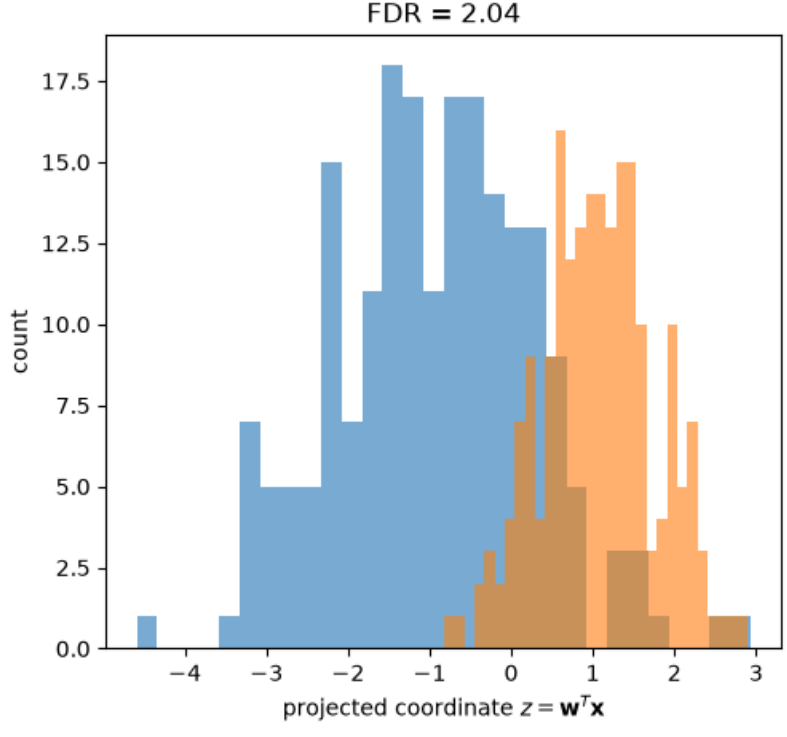
$$\implies \mathbf{w} \propto \boldsymbol{S}_w^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$

II. If $P(w_1) \neq P(w_2)$:

$$\mathbf{w} = \text{leading eigenvector of } \boldsymbol{S}_w^{-1}\boldsymbol{S}_B$$

Projection angle (degrees)


Slider direction w vs Fisher optimum w_{opt}
w
w_{opt}
x_1
x_2


FDR = 2.04
count
projected coordinate $z = \mathbf{w}^T\mathbf{x}$

Remark

- Generalized to $\mathbf{z} = \mathbf{W}^T\mathbf{x} \in \mathbb{R}^k$ where $k \leq d$ and $\mathbf{W} \in \mathbb{R}^{d \times k}$.
    - More complex (pages 291-297 in book).

Principal Component Analysis (PCA)

- First: $\mathbf{z} = \mathbf{A}\mathbf{x}$ such that $\mathbf{z} \in \mathbb{R}^d$, $\mathbf{x} \in \mathbb{R}^d$, and $\mathbf{A} \in \mathbb{R}^{d \times d}$
- Want: $\mathbf{\Sigma}_z$ diagonal!

A closer look at the covariance matrix

- Have:

$$\boldsymbol{\Sigma}_z = \mathbb{E}[(\mathbf{z} - \boldsymbol{\mu}_z)(\mathbf{z} - \boldsymbol{\mu}_z)^T]$$

$$= \mathbb{E}[(\mathbf{A}\mathbf{x} - \mathbf{A}\boldsymbol{\mu}_x)(\mathbf{A}\mathbf{x} - \mathbf{A}\boldsymbol{\mu}_x)^T] = \mathbf{A}\,\boldsymbol{\Sigma}_x\mathbf{A}^T$$

- $\boldsymbol{\Sigma}_x$: symmetric and positive semi-definite $\implies$ orthogonal eigenvectors and non-negative eigenvalues.

Eigendecomposition of the covariance matrix

- Let $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_d]$

$$\boldsymbol{\Sigma}_x \mathbf{E} = \mathbf{E} \boldsymbol{\Lambda}$$

where

$$\boldsymbol{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_d \end{bmatrix}$$

(diagonal matrix of eigenvalues)

- With $\mathbf{A} = \mathbf{E}^T$ (orthonormal), the covariance of the transformed data is diagonal:

Interpreting the eigenvalues and eigenvectors

- $\boldsymbol{\Sigma}_z = \boldsymbol{\Lambda}$: the transformed features are uncorrelated.
- The variance of z_i equals λ_i.
- The eigenvectors $\mathbf{e}_i$ are the directions of maximal variance in the original space.

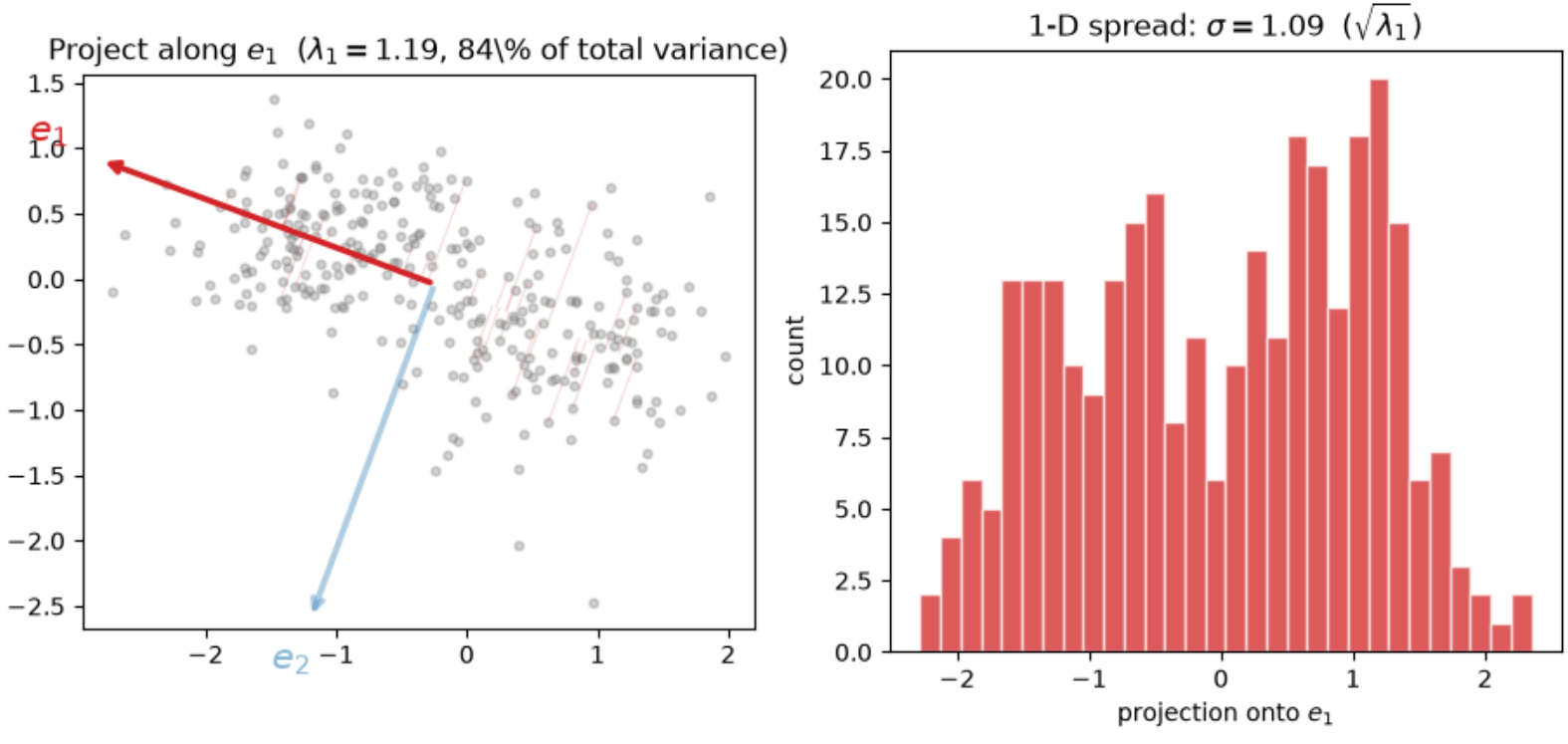
Variance maximally preserved

- First:
  $\sum_{i=1}^{d} \mathrm{Var}(z_i) = \mathrm{trace}(\mathbf{\Sigma}_z) =$
  $\sum_{i=1}^{d} \lambda_i$
- Thus: Let $\mathbf{A} = \mathbf{E}^T = [\mathbf{e}_1, \ldots, \mathbf{e}_d]^T$
- Remark: assume $\mathbb{E}[\mathbf{x}] = 0$ (center the data first).

Example: projecting onto the eigenvectors

- The data cloud below is **unlabeled** — PCA only sees the point cloud.
- Use the dropdown to project onto e_1 (most variance) or e_2 (least).
- Compare the spread of the 1-D projections with $\sqrt{\lambda_i}$.

Project along e_1 (most variance)


Project along e_1 ($\lambda_1 = 1.19$, 84\% of total variance)
e_1
e_2
1-D spread: $\sigma = 1.09$ ($\sqrt{\lambda_1}$)
count
projection onto e_1

PCA is reconstruction / compression

- Let $\mathbf{z} \in \mathbb{R}^d = [z_{(0)}, z_{(1)}, \ldots, z_{(d-1)}]^T$

PCA is reconstruction / compression

- Have $\mathbf{x} = \mathbf{A}^T\mathbf{z}$ (with $\mathbf{A} = \mathbf{E}^T$)

PCA is reconstruction / compression

- **MSE:**
  $\mathbb{E}\left[\|\mathbf{x} - \hat{\mathbf{x}}\|^2 \right]$

For $\mathbf{z} \in \mathbb{R}^k$: $\hat{\mathbf{x}} = \sum_{i=0}^{k-1} z(i)\mathbf{e}_i$

If $z(i) = 0$ for $i \geq k$:

$$\mathbb{E}\left[\|\mathbf{x} - \hat{\mathbf{x}}\|^2\right] = \sum_{i=k}^{d-1} \lambda_i$$

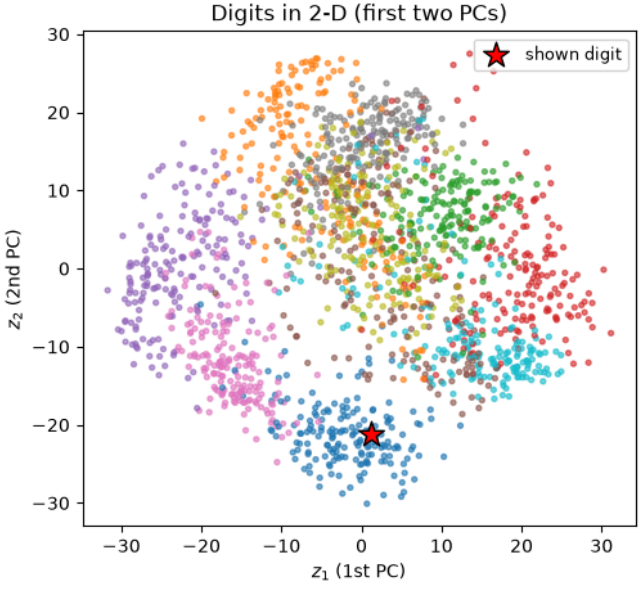
$\Longrightarrow$ **MSE is minimized**.

- **Compression:** Store/save $\mathbf{z} \in \mathbb{R}^k$ instead of $\mathbf{x}$ (e.g. images).
- **Reconstruct:** $\hat{\mathbf{x}}$ using $\mathbf{z}$.

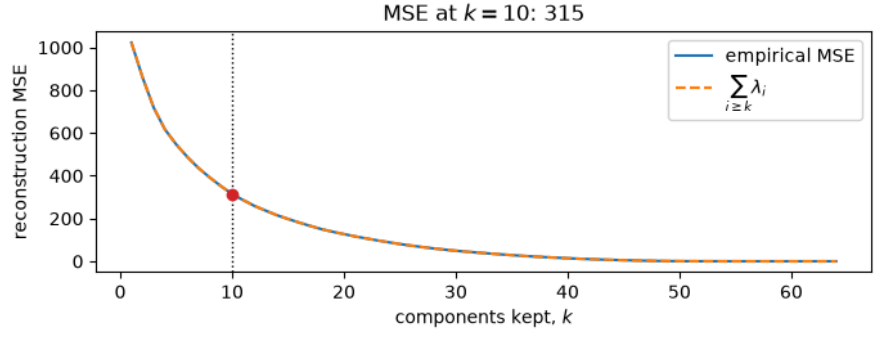
Example: compressing handwritten digits

- Each 8×8 digit image is a point in $\mathbb{R}^{64}$.
- Drag the slider to keep the first k principal components and reconstruct.
- The reconstruction error should follow $\sum_{i \geq k} \lambda_i$.

Principal components kept, k


Digits in 2-D (first two PCs)
shown digit
z_2 (2nd PC)
z_1 (1st PC)
30
20
10
0
−10
−20
−30
−30
−20
−10
0
10
20
30


Original
Reconstruction, $k = 10$


MSE at $k = 10$: 315
empirical MSE
$\sum_{i \geq k} \lambda_i$
reconstruction MSE
components kept, k
1000
800
600
400
200
0
0
10
20
30
40
50
60